Marketing Materials

Blue Sea Systems offers sales and marketing materials that assist in the merchandising, promotion, and selection of products and the Blue Sea Systems brand. A selection of materials are shown on this page.

For updated information and new marketing and sales materials, visit www.bluesea.com/resources/134.

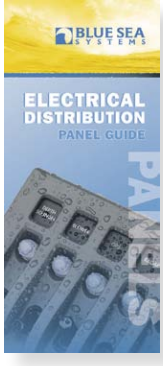


20003
Blue Sea Systems Hat
- Heavyweight 100% brushed cotton
- Adjustable strap
- Navy blue
- One size

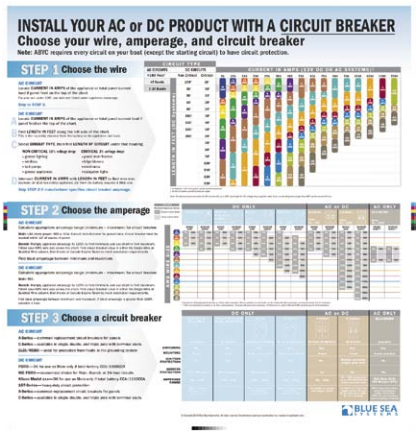
20004
Blue Sea Systems Hat
- Heavyweight 100% brushed cotton
- Adjustable strap
- Stone
- One size



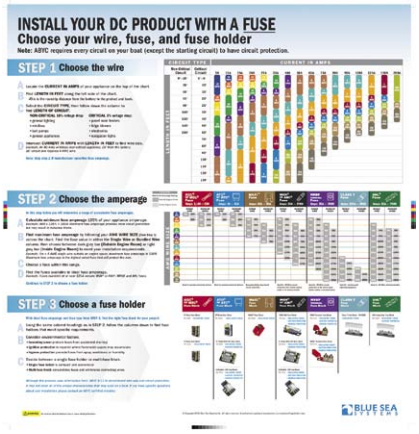
20007
2011 Blue Sea Systems Catalog
- 124 pages
- 36 catalogs in a case
- Order individually



20001
Blue Sea Systems Panel Guide
- Consumer marketing tool
- Outlines above and below deck panels and accessories
- 20 guides per pack



20005
Circuit Breaker Selection Sign
- Assists in determining the correct circuit breaker for the application
- Full color
- Heavyweight PVC material
- 24" wide x 24" high
- Order individually



20006
Fuse and Fuse Block Selection Sign
- Assists in determining the correct fuse and fuse block for the application
- Full color
- Heavyweight PVC material
- 24" wide x 24" high
- Order individually



Merchandising Plans
- Blue Sea Systems offers plan-o-grams to assist with merchandising products.
- The plans are available in various sizes and formats.
- Updated plans are always available at www.bluesea.com/resources/134.





20002 Front Back
VSM 422 Hang Tag
- Showcases the features and benefits of the VSM 422
- Card merchandises on two pegs
- 20 cards per pack

QUICK GUIDE TO CIRCUIT BREAKERS

Thermal Circuit Breakers

AC Amperage Range | DC Amperage Range



Page 32

Push Button Reset-Only—Appropriate for 24-hour circuit protection

Iic Interrupting Capacity: 3,000A @ 14.7V DC/2,500A @ 28V DC Vmxo Maximum Voltage: 32V DC

IP Meets ignition protection requirements

Itr Available amperages
3, 5, 7, 10, 15, 20, 25, 30, 40

3–40A



Page 33

Medium Duty Push Button Reset-Only—Appropriate for 24-hour circuit protection

Iic Interrupting Capacity: 5,000A @ 32V DC Vmxo Maximum Voltage: 32V DC

IP Meets ignition protection requirements

Itr Available amperages
15, 20, 30, 40, 50, 60

15–60A



Page 34

285-Series—Appropriate for DC Main circuit protection with battery banks under 1,100 CCA

Iic Interrupting Capacity: 3,000A DC @ 12V DC Vmxo Maximum Voltage: 24V DC

IP Meets ignition protection requirements

Itr Available amperages
25, 30, 40, 50, 60, 70, 80, 100, 120, 150, 200

25–200A



Page 35

187-Series—Appropriate for DC Main circuit protection with battery banks over 1,100 CCA

Iic Interrupting Capacity: 5,000A @ 12V DC / 3,000A @ 24V DC/1,500A@42 DC Vmxo Maximum Voltage: 48V DC

IP Meets ignition protection requirements

Itr Available amperages
25, 30, 35, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 200A

25–200A

Magnetic Hydraulic Circuit Breakers



Page 36

A-Series Toggle, 1 Pole—Appropriate for AC and DC Branch circuit protection

Iic Interrupting Capacity: 7,500A @ 65V DC/3,000A @ 120V AC/3,000A @ 250V AC Vmxo Maximum Voltage: 65V DC/250V AC

Itr Available amperages
5, 8, 10, 15, 20, 25, 30, 40, 50

5–50A



Page 36

A-Series Toggle, 2 Pole—Appropriate for 120V AC Main or 240V AC Branch circuit protection

Iic Interrupting Capacity: 3,000A @ 120V AC/3,000A @ 250V AC Vmxo Maximum Voltage: 250V AC

Itr Available amperages
10, 15, 16, 20, 30, 32, 40, 50

10–50A



Page 37

A-Series Flat and Restricted OFF Rocker, 1 Pole—Appropriate for AC and DC Branch and 24-hour circuit protection

Iic Interrupting Capacity: 5,000A @ 32V DC/3,000A @ 120V AC/1,500A @ 250V AC Vmxo Maximum Voltage: 32V DC/250V AC

Itr Available amperages
5, 8, 10, 15, 20, 25, 30, 40, 50

5–50A



Page 37

A-Series Flat and Raised Rocker, 2 Pole—Appropriate for 120V AC Main or 240V AC Branch circuit protection

Iic Interrupting Capacity: 5,000A @ 32V DC/3,000A @ 240V AC Vmxo Maximum Voltage: 240V AC

Itr Available amperages
10, 15, 16, 20, 30, 32, 40, 50

10–50A



Page 38

C-Series Toggle, 1 Pole—Appropriate for DC Main and AC and DC high load circuit protection

Iic Interrupting Capacity: 10,000A @ 80V DC/5,000A @ 250V AC Vmxo Maximum Voltage: 80V DC/250V AC

IP (7250I ONLY) Meets ignition protection requirements

Itr Available amperages
5, 10, 15, 20, 25, 30, 50, 60, 80, 100

5–100A



Page 38

C-Series Toggle, 2 and 3 Parallel Pole—Appropriate for DC high load circuit protection

Iic Interrupting Capacity: 5,000A @ 65V DC Vmxo Maximum Voltage: 65V DC

Itr Available amperages
150, 175, 200, 250, 300

150–300A



Page 38

C-Series Toggle, 2 and 3 Pole—Appropriate for 240V AC Main and AC high load circuit protection

Iic Interrupting Capacity: 5,000A @ 250V AC Vmxo Maximum Voltage: 250V AC

Itr Available amperages
30, 50, 60, 80, 100

30–100A



Page 39

C-Series Flat Rocker, 1 Pole—Appropriate for DC Main and AC and DC high load circuit protection

Iic Interrupting Capacity: 5,000A @ 32V DC/3,500A @ 240V AC Vmxo Maximum Voltage: 32V DC/240V AC

IP Meets ignition protection requirements

Itr Available amperages
5, 10, 15, 20, 25, 30, 50, 60, 80, 100

5–100A



Page 39

C-Series Flat Rocker, 2 and 3 Parallel Pole—Appropriate for DC high load circuit protection

Iic Interrupting Capacity: 5,000A @ 48V DC Vmxo Maximum Voltage: 48V DC

Itr Available amperages
150, 175, 200, 250, 300

150–300A



Page 39

C-Series Flat and Raised Rocker, 2 and 3 Pole—Appropriate for 240V AC Main and AC high load circuit protection

Iic Interrupting Capacity: 5,000A @ 240V AC Vmxo Maximum Voltage: 240V AC

Itr Available amperages
30, 50, 60, 80, 100

30–100A



Page 41

Residual Current (RCBO—includes GFCI and ELCI), 1 and 2 Pole—Appropriate for ground fault and overcurrent trip protection

Iic Interrupting Capacity (I_{ic}): 5,000A AC Vmxo Maximum Voltage: 240V AC

Itr Available amperages
15, 30, 50

15-50A

QUICK GUIDE TO FUSES, FUSE HOLDERS AND FUSE BLOCKS

FUSES

DC Amperage Range



Page 43

AGC® and MDL® Fuse—Appropriate for small electronic devices
Iic Interrupting Capacity: 1,000A DC Vmxo Maximum Voltage: 32V DC
.25-30A
Itr Available amperages
.25, .5, 1, 1.5, 2, 2.5, 3, 4, 5, 6.25, 7, 7.5, 10, 15, 20, 25, 30



Page 43

GMA® and AGA® Fuse—Appropriate for small electronic devices
Iic Interrupting Capacity: 1,000A DC Vmxo Maximum Voltage: 32V DC
1-20A
Itr Available amperages
1, 2, 3, 4, 5, 7, 10, 20



Page 43

ATO® or ATC® Fuse—Appropriate for small electronic devices
Iic Interrupting Capacity: 1,000A DC Vmxo Maximum Voltage: 32V DC
1-30A
Itr Available amperages
1, 2, 3, 4, 5, 7.5, 10, 15, 20, 25, 30



Page 43

ATM® Fuse—Appropriate for small electronic devices
Iic Interrupting Capacity: 1,000A DC Vmxo Maximum Voltage: 32V DC
5-30A
Itr Available amperages
5, 10, 15, 20, 30



Page 43

MAXI™ Fuse—Economical high amp branch circuit protection
Iic Interrupting Capacity: 1,000A DC Vmxo Maximum Voltage: 32V DC
30-80A
Itr Available amperages
30, 40, 50, 60, 70, 80



Page 44

MEGA® or AMG® Fuse—Appropriate for DC Main circuit protection with smaller battery banks or DC Branch circuits
Iic Interrupting Capacity: 2,000A DC Vmxo Maximum Voltage: 32V DC
100-300A
Itr Available amperages
100, 125, 150, 175, 200, 225, 250, 300



Page 44

MIDI® or AMI® Fuse—Appropriate for DC Main or DC Branch circuit protection
Iic Interrupting Capacity: 5,000A DC Vmxo Maximum Voltage: 32V DC
30-200A
Itr Available amperages
30, 40, 50, 60, 70, 80,100, 125, 150, 175, 200



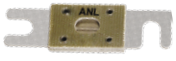
Page 44

Terminal Fuse—Appropriate for DC Main circuit protection at the battery post, battery switch, or terminal block
Iic Interrupting Capacity: 10,000A @ 14V DC/5,000A @ 32V DC/2,000A @ 58V DC Vmxo Maximum Voltage: 58V DC
IP Meets Ignition Protection requirements
30-300A
Itr Available amperages
30, 40, 50, 60, 75, 80, 90, 100, 125, 150, 175, 200, 225, 250, 300



Page 45

Class T Fuse—Appropriate for inverters and high amp equipment
Iic Interrupting Capacity: 20,000A DC Vmxo Maximum Voltage: 160V DC
225-400A
Itr Available amperages
225, 250, 300, 350, 400



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ANL® Fuse—Appropriate for DC Main circuit protection
Iic Interrupting Capacity: 6,000A DC Vmxo Maximum Voltage: 32V DC
IP 35-500 Ampere Fuses meet Ignition Protection requirements
Available amperages (I_{tr})
35, 40, 50, 60, 80, 100, 130, 150, 175, 200, 225, 250, 275, 300, 325, 350, 400, 500, 600, 675, 750
35-750A

FUSE HOLDERS



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Crimpable Fuse Holder—For use with AGC® or MDL® glass fuses
Imxo Maximum Amperage per circuit: 30A DC



Page 46

AGC® Waterproof Fuse Holder—For use with AGC® or MDL® glass fuses
Imxo Maximum Amperage per circuit: 30A DC



Page 46

AGC® Waterproof Fuse Holder—For use with AGC® or MDL® glass fuses
Imxo Maximum Amperage per circuit: 20A DC



Page 46

AGC® or MDL® In-Line Fuse Holder—Heavy duty fuse holder for AGC® or MDL® glass fuses
Imxo Maximum Amperage per circuit: 30A DC



Page 46

ATO® or ATC® In-Line Fuse Holder
Imxo Maximum Amperage per circuit: 30A DC



ATO® or ATC® Waterproof Fuse Holder
Imxo Maximum Amperage per circuit: 30A DC

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FUSE BLOCKS



ST Glass Fuse Block (6 Circuit Models Available)—Uses AGC® or MDL® Fuses
Vmxo Maximum Voltage: 32V DC Imxo Maximum Amperage per circuit: 30A DC Imxo Maximum Amperage per block: 100A DC

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MAXI™ Fuse Block—Uses MAXI™ Fuses
Vmxo Maximum Voltage: 32V DC Imxo Maximum Amperage: 80A DC

Page 46



ST Blade Fuse Block (6 and 12 Circuit Models Available)—Uses ATO® or ATC® Fuses
Vmxo Maximum Voltage: 32V DC Imxo Maximum Amperage per circuit: 30A DC Imxo Maximum Amperage per block: 100A DC

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Terminal Fuse Block—Uses Terminal Fuses
Vmxo Maximum Voltage: 58V DC Imxo Maximum Amperage: 300A DC

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MIDI® or AMI® Safety Fuse Block—Uses MIDI® or AMI® fuses
Vmxo Maximum Voltage: 58V DC Imxo Maximum Amperage: 300A DC

Page 48



MEGA® or AMG® Safety Fuse Holder—Uses MEGA® or AMG® fuses
Vmxo Maximum Voltage: 58V DC Imxo Maximum Amperage: 300A DC

Page 48



Class T Fuse Block—Uses Class T Fuses
Vmxo Maximum Voltage: 160V DC Imxo Maximum Amperage: 400A DC

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ANL® Fuse Block—Uses ANL® Fuses
Vmxo Maximum Voltage: 32V DC
Imxo Maximum Amperage: 300A DC



ANL® Heavy Duty Fuse Block—Uses ANL® Fuses
Vmxo Maximum Voltage: 32V DC
Imxo Maximum Amperage: 750A DC

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SafetyHub 100 Fuse Block—Holds 3 MIDI® or AMI® and 4 ATO® or ATC® Fuses
Vmxo Maximum Voltage: 12V DC Imxo Maximum Amperage per block: 300A DC
MIDI® or AMI® Fuse Block—Imxo Maximum Amperage per circuit: 200A DC
ATO® or ATC® Fuse Block—Imxo Maximum Amperage per circuit: 30A DC Imxo Maximum Amperage per block: 50A DC

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SafetyHub 150 Fuse Block—Holds 4 MIDI® or AMI® and 6 ATO® or ATC® Fuses
Vmxo Maximum Voltage: 12V DC Imxo Maximum Amperage per block: 350A DC
MIDI® or AMI® Fuse Block—Imxo Maximum Amperage per circuit: 200A DC
ATO® or ATC® Fuse Block—Imxo Maximum Amperage per circuit: 30A DC Imxo Maximum Amperage per block: 50A DC

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SafetyHub 250 Fuse Block with Remote Battery Switch—Holds 3 MIDI® or AMI® and 4 ATO® or ATC® Fuses
Vmxo Maximum Voltage: 12V DC Imxo Maximum Amperage: 240A DC
MIDI® or AMI® Fuse Block—Imxo Maximum Amperage per circuit: 200A DC
ATO® or ATC® Fuse Block—Imxo Maximum Amperage per circuit: 20A DC Imxo Maximum Amperage per block: 50A DC

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DC Main Circuit Protection and Branch Circuit Protection

Purpose

Fuses and circuit breakers are used to protect wire insulation from melting and starting fires in the event of overcurrents or short circuits which cause more amperage to flow in a wire than that wire is rated to carry. It is important to note that, except for those wires that are intended to carry starting currents, *every positive wire* in the DC Main Power Distribution System must be protected by a fuse or circuit breaker.

Considerations for DC Main Circuit Protection

Mounting Placement–distance from power source.

The DC Main circuit protection system uses circuit breakers or fuses to protect the wires of the DC Main distribution system. The American Boat and Yacht Council (ABYC) publishes voluntary standards for the type and placement of the fuse or circuit breaker to be used as a DC Main circuit protection device. Wire intended to carry engine starting currents between the batteries, the switch, and the starter is not required to have main circuit protection devices installed. Maximum mounting placement dimensions for a fuse or circuit breaker are 7" if the conductor is not housed in a sheath or enclosure in addition to the wire insulation, 40" if the conductor is housed in a sheath or enclosure in addition to the wire insulation, and 72" if the conductor is connected directly to the battery and housed in a sheath or enclosure in addition to the wire insulation.

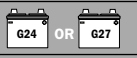
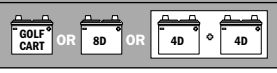
Selecting DC Main Circuit Protection.

The principal attribute of a DC Main circuit protection device is its Ampere Interrupt Capacity (AIC) rating. Specifications listed in the ABYC standards determine the AIC a DC Main circuit protection device must have. The required AIC rating is determined by the total CCA of the batteries connected to the circuit. See the tables at right for the required AIC ratings.

Wire selection for DC applications on boats is usually based on voltage drop requirements. However, there is a maximum continuous current that the wire can withstand without overheating. Higher grade marine wires are rated for service up to 105°C (221°F)—the ABYC wire capacity table for 105°C is most frequently quoted. The 105°C table accurately reflects the capacity of single conductors exposed to freely circulating cooling air. However, other factors, such as covering bundles of wire in outer jackets to form a cable, or use of conduits or structural voids to protect wires, can reduce the cooling and reduce the safe capacity of the wire. A more conservative strategy is to use the 105°C wire, but treat it according to the 75°C table above when selecting circuit protection unless the wire is openly exposed for cooling.

See the Blue Sea Systems Circuit Wizard on our Web site at circuitwizard.bluesea.com for more assistance with wire and circuit protection selection.

ABYC Interrupt Rating Table

Total Connected Battery Cold Cranking Amperes (CCA) *		Ampere Interrupt Capacity	
12 VOLTS AND 24 VOLTS			
The white boxes identify two batteries, of the same size, placed in parallel configuration.		DC MAIN	DC BRANCH
	650 CCA or Less	1,500 AIC	750 AIC
	651–1,100 CCA	3,000 AIC	1,500 AIC
	Over 1,100 CCA	5,000 AIC	2,500 AIC
32 VOLTS			
	1,250 CCA or Less	3,000 AIC	1,500 AIC
	Over 1,250 CCA	5,000 AIC	2,500 AIC

* Battery cold cranking performance rating at -17.8°C (0°F): The discharge load in amperes that a battery at -17.8°C (0°F) can deliver for 30 seconds, and maintain a voltage of 1.2 Volts per cell or higher, (e.g. 7.2 Volts for a 12 Volt battery). The CCA for the battery icons in this chart is an approximation and could be slightly higher or lower. Consult the battery with manufacturer's specifications for precise CCA ratings. A battery rated in MCA will have a CCA capacity approximately 80% of MCA

ABYC E-11 requires the use of circuit breakers that can be reused and reset and that they be applied as per the table above. The standard does not strictly require that fuses be applied in the same way, but it is an issue to consider, especially with high amperage fuses used to protect panel feeders or inverters. Fuses under 10 Ampere rating generally have such a high internal resistance they prevent fault currents from reaching 1000 Amperes in 12 Volt circuits. The apparent contradiction when using these fuses for bilge pumps and other circuits directly off the battery is less of an issue than it might seem. If a fuse blows, and the case appears to be cracked or metal has been ejected, the fuse holder should be replaced.

ABYC Ampacity Rating Table at 30°C

WIRE SIZE		TEMPERATURE RATING OF CONDUCTOR INSULATION												REFERENCE DATA		
US	Metric	75°C		90°C		105°C		75°C		90°C		105°C			Ohms	Ohms
AWG	mm^2		EngRm		EngRm		EngRm		EngRm		EngRm		EngRm	mm dia	/1000ft	/1000m
	0.75	9.5	7	19	15.5	19	16	6.6	5.0	13	11	13	11	0.98	7.29	23.92
18	0.82	10	8	20	16	20	17	7	5	14	12	14	12	1.02	6.67	21.88
	1.0	13	10	21	17	21	18	9	7	15	12	15	13	1.13	5.47	17.94
16	1.3	15	11	25	21	25	21	11	8	18	14	18	15	1.29	4.17	13.70
	1.5	16	12	24	20	29	24	11	9	17	14	20	17	1.38	3.65	11.96
14	2.1	20	15	30	25	35	30	14	11	21	17	25	21	1.63	2.63	8.63
	2.5	21	16	34	28	38	32	15	11	23	19	26	22	1.78	2.19	7.18
12	3.3	25	19	40	33	45	38	18	13	28	23	32	27	2.05	1.65	5.42
	4.0	34	25	46	38	51	43	24	18	32	27	35	30	2.26	1.37	4.49
10	5.3	40	30	55	45	60	51	28	21	39	32	42	36	2.59	1.04	3.41
	6.0	53	40	57	47	65	55	37	28	40	33	45	39	2.76	0.91	2.99
8	8.4	65	49	70	57	80	68	46	34	49	40	56	48	3.27	0.65	2.14
	10.0	79	60	84	69	100	85	56	42	59	48	70	60	3.6	0.55	1.79
6	13.3	95	71	100	82	120	102	67	50	70	57	84	71	4.1	0.41	1.35
	16.0	105	79	113	93	134	114	73	55	79	65	94	80	4.5	0.34	1.12
4	21	125	94	135	111	160	136	88	66	95	78	112	95	5.2	0.26	0.85
	25	141	106	150	123	175	148	99	74	105	86	122	104	5.6	0.22	0.72
3	27	145	109	155	127	180	153	102	76	109	89	126	107	5.8	0.21	0.67
2	34	170	128	180	148	210	179	119	89	126	103	147	125	6.5	0.16	0.53
	35	173	130	186	153	217	185	121	91	130	107	152	129	6.7	0.16	0.51
1	42	195	146	210	172	245	208	137	102	147	121	172	146	7.3	0.13	0.42
	50	220	165	235	193	273	232	154	116	164	135	191	163	8.0	0.109	0.36
0	54	230	173	245	201	285	242	161	121	172	141	200	170	8.3	0.102	0.34
00	68	265	199	285	234	330	281	186	139	200	164	231	196	9.3	0.081	0.27
	70	274	206	292	239	341	289	192	144	204	168	238	203	9.4	0.078	0.26
000	85	310	233	330	271	385	327	217	163	231	189	270	229	10.4	0.064	0.21
	95	334	251	357	293	413	351	234	175	250	205	289	246	11.0	0.058	0.19
0000	107	360	270	385	316	445	378	252	189	270	221	312	265	11.7	0.051	0.17
	120	387	290	414	339	478	406	271	203	290	237	335	284	12.4	0.046	0.15
	150	445	333	476	390	550	467	311	233	333	273	385	327	13.8	0.036	0.12

Data based on E-11 Table VI-A (Single Conductors in Free Air)

Data based on E-11 Table VI-B (Up to three conductors in a sheath, conduit or bundle)

SAE conductors are smaller than equivalent AWG by 5% to 12% with current capacity typically less by 7%. ISO Ratings for metric wire are slightly less than these values derived from ABYC VI-A ratings.

- For bundles of 4 to 6 conductors multiply by 0.857
- For bundles of 7 to 24 conductors multiply by 0.714
- For bundles of 25 or more, conductors multiply by 0.571

Wires counted in bundles need not include:

1. Wires carrying intermittent currents no more than rating per VI-A and for less than one minute per mm of diameter, and not repeating more often than a delay of 5X times active duration.
2. Wires carrying load currents at less than 50% of the wire rating per table VI-B.

† Thermally limited amperage capacity

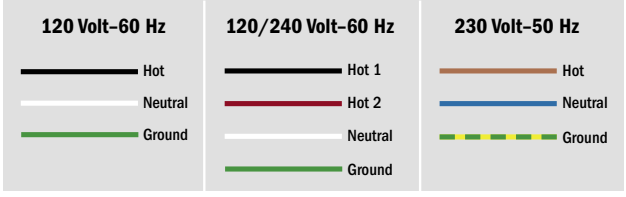
AC Main Power Distribution and Circuit Protection

Purpose

- Provide a path for delivering power from the ship's sources of AC power to the AC branch distribution system
- Provide a path for returning fault currents to ground via the green safety ground wire
- Provide a means for disconnecting AC power when the boat is not in use or in emergencies
- Provide electrical separation to insure that two sources of AC power are never connected
- Provide circuit protection for neutral and line wires in the AC main system
- Provide ground fault protection

AC Wire Systems

The three most common AC systems used on boats are shown here. In all cases the ground, sometimes called safety ground to clarify its purpose and differentiate it from the DC ground or negative, is said to be a "normally non-current carrying wire." Its purpose is to provide the lowest resistance path for AC currents that have strayed from their proper containment in the normally current carrying hot and neutral wires. The ground wire is connected to the exterior conductive parts of AC devices that could be touched by a person during normal operation, and it conducts errant AC currents safely to ground rather than passing them through a human body. The ground wire is never passed through a circuit breaker.



Devices Qualifying as AC Main Circuit Breakers

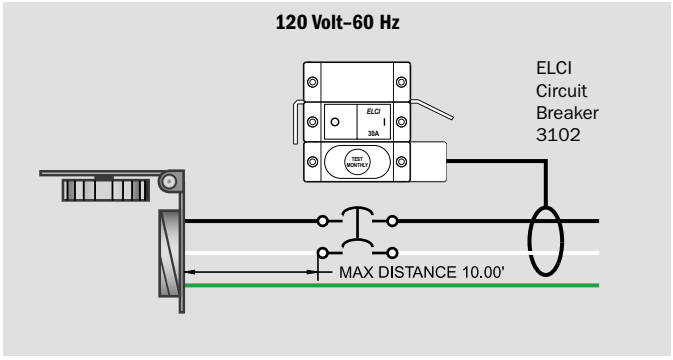
In order to qualify as an AC main circuit breaker, these characteristics must be present:

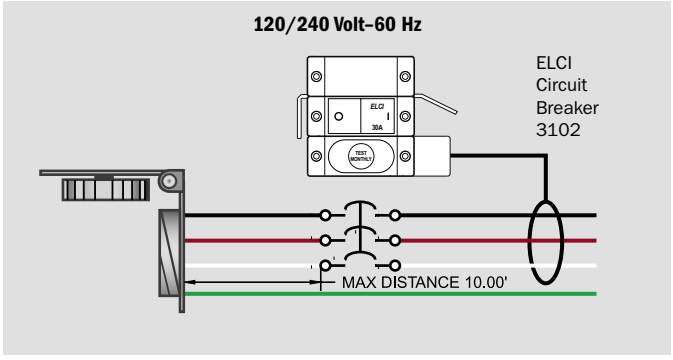
1. The circuit breaker must have an Ampere Interrupt Capacity (AIC) meeting the requirements of the following tables.
2. The circuit breaker must be multiple pole, usually 2 or 3 (see AC Wire Systems).
3. The circuit breaker must be rated for the appropriate AC system voltage in which it will be used.
4. The circuit breaker must be available in amperages appropriate to the design amperage of the system. In the USA, this is generally 30A and 50A, while European systems are generally 16A and 32A.
5. The ELCI shall have a leakage trip mechanism that trips below 30mA when current leaks to ground.

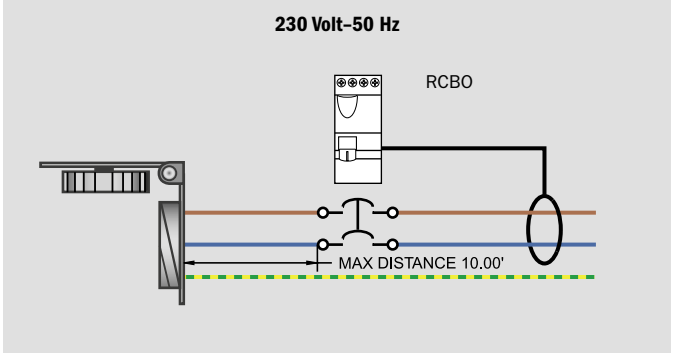
AC Shore Power Source	Main Circuit Breaker	Branch Circuit Breaker
120V - 30A	3,000	3,000
120V - 50A	3,000	3,000
120/240V - 50A	5,000	3,000
240V - 50A	5,000	3,000

Sources of AC power, whether shore power or onboard generators and inverters, should always have a circuit breaker near the power source. This circuit breaker is designated the AC main circuit breaker. The AC main circuit breaker should always have a pole for each of the hot and neutral wires in the circuit assuring that circuit protection functions are not compromised in reverse polarity situations.

Beginning in July 2010 ABYC Standards require that an Equipment Leakage Circuit Interrupter (ELCI) with a 30mA leakage trip be installed in shore power applications as the first protective device after the power inlet. ELCIs respond to leakage of electrical current outside of the intended current path, and provide overload and short circuit protection. They serve as the main AC circuit breaker for the system. These devices will open all energized conductors and the neutral when opened manually or tripping on an overload or leakage fault. For a more complete discussion of ELCIs, see page 40–41.







Choosing The Correct Wire

A Locate the CURRENT IN AMPS of your appliance across the top of the chart. Most electrical products include a rating label, or you can find the amperage rating in the documentation that came with the product.

B Find circuit LENGTH IN FEET along the left side of the chart. Note that the total length of the circuit is the roundtrip distance from power source (usually the battery) to the product and back to the source.

C Select the CIRCUIT TYPE. Allowable voltage drop is based on whether a circuit is critical or non-critical.

Critical circuits, with 3% allowable voltage drop, include

- Panel main feeders
- Bilge blowers
- Electronics
- Navigation lights

Non-critical circuits, with 10% allowable voltage drop, include

- General lighting
- Windlasses
- Bait pumps
- General appliances

Follow down the column until you find your circuit's LENGTH IN FEET.

D Intersect CURRENT IN AMPS with LENGTH IN FEET to identify the wire size.

Example: A windlass rated 80A is 25' from the battery. Circuit length is 50', circuit type is 'non-critical', and correct wire size is 4 AWG.

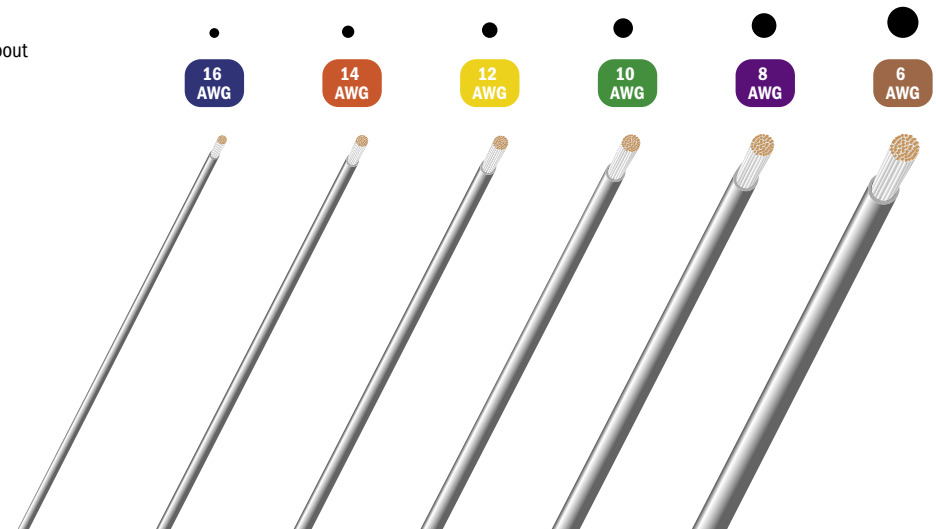
Although this process uses information from ABYC E-11 to recommend wire size, it may not cover all of the unique characteristics that exist on a boat. If you have specific questions about your installation, please consult an ABYC certified installer.

Wire Selection Chart

CIRCUIT TYPE						
LENGTH IN FEET	10% VOLTAGE DROP Non-Critical Circuit	3% VOLTAGE DROP Critical Circuit	5A	10A	15A	20A
	0'–20'	0'–6'	16 AWG	16 AWG	14 AWG	14 AWG
	30'	10'		14 AWG	12 AWG	12 AWG
	50'	15'		12 AWG	10 AWG	10 AWG
	65'	20'	14 AWG	10 AWG		8 AWG
	80'	25'	12 AWG		8 AWG	6 AWG
	100'	30'		8 AWG		
	130'	40'	10 AWG		6 AWG	4 AWG
	165'	50'		6 AWG		
	200'	60'			4 AWG	2 AWG
		70'	8 AWG			
		80'		4 AWG	2 AWG	
		90'				1 AWG
		100'	6 AWG			
		110'		2 AWG		
		120'				
		130'				

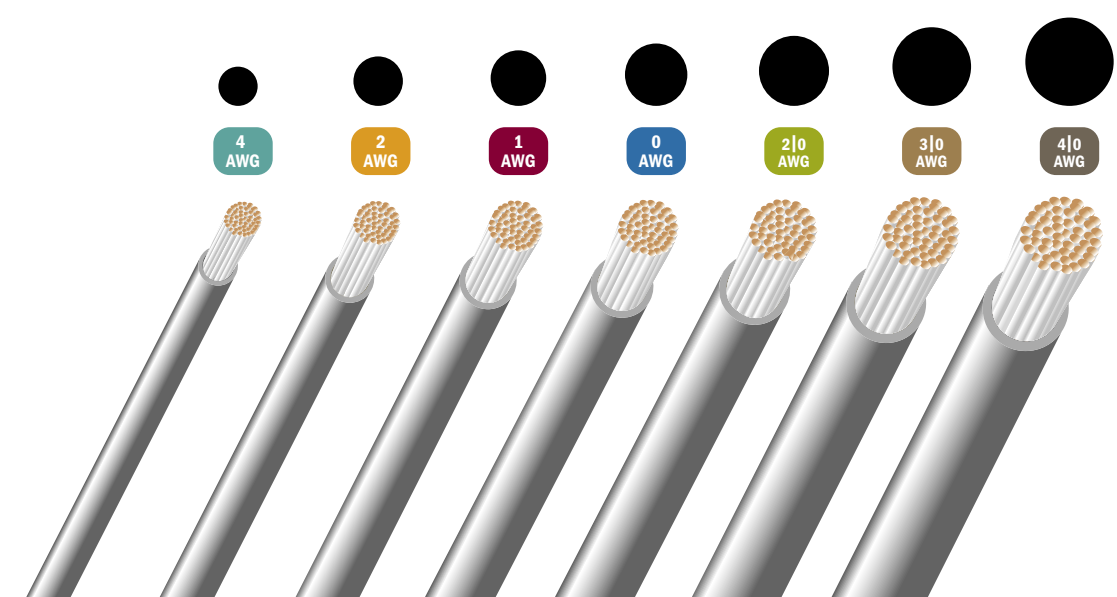
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Black circles indicate actual diameter of conductor (not insulation).



CURRENT IN AMPS

25A	30A	40A	50A	60A	70A	80A	90A	100A	120A	150A	200A
12 AWG	10 AWG	8 AWG	6 AWG	6 AWG	6 AWG	4 AWG	4 AWG	4 AWG	2 AWG	1 AWG	2|0 AWG
10 AWG	8 AWG	6 AWG	4 AWG	4 AWG	4 AWG	2 AWG	2 AWG	2 AWG	1 AWG	0 AWG	3|0 AWG
8 AWG	6 AWG	4 AWG	2 AWG	2 AWG	2 AWG	1 AWG	1 AWG	1 AWG	0 AWG	2|0 AWG	4|0 AWG
6 AWG	4 AWG	2 AWG	1 AWG	1 AWG	1 AWG	0 AWG	0 AWG	0 AWG	2|0 AWG	3|0 AWG	
4 AWG	2 AWG	1 AWG	0 AWG	0 AWG	0 AWG	3|0 AWG	2|0 AWG	2|0 AWG	3|0 AWG	4|0 AWG	
2 AWG	1 AWG	0 AWG	2|0 AWG	2|0 AWG	2|0 AWG	4|0 AWG	3|0 AWG	3|0 AWG	4|0 AWG		
1 AWG	0 AWG	2|0 AWG	3|0 AWG	3|0 AWG	3|0 AWG		4|0 AWG	4|0 AWG			
0 AWG	2|0 AWG	3|0 AWG	4|0 AWG	4|0 AWG	4|0 AWG						



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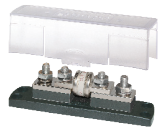

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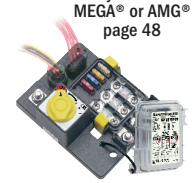

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